

11 ABC is a right-angled triangle.

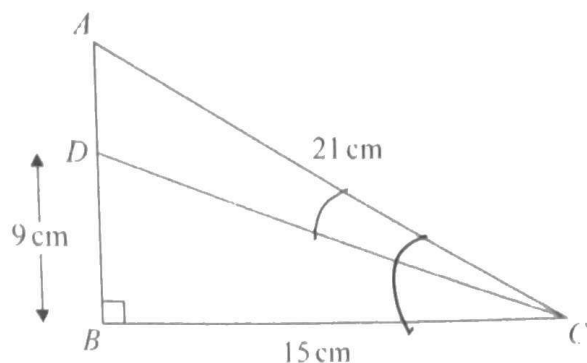


Diagram NOT
accurately drawn

$$AC = 21 \text{ cm} \quad BC = 15 \text{ cm} \quad \text{angle } ABC = 90^\circ$$

The point D lies on AB such that $DB = 9 \text{ cm}$

Work out the size of angle ACD

Give your answer correct to one decimal place.

Pythagoras theorem,

$$AC^2 = AD^2 + BC^2$$

$$21^2 = AD^2 + 15^2$$

$$AD^2 = 21^2 - 15^2$$

$$AD = \sqrt{21^2 - 15^2} = 6\sqrt{6}$$

$\triangle ACB$

$$\tan \theta = \frac{6\sqrt{6}}{15}$$

$$\theta = \tan^{-1}\left(\frac{6\sqrt{6}}{15}\right)$$

$$\theta = 44.415$$

$$\begin{aligned} \therefore \hat{ACD} &= \hat{ACB} - \hat{DCB} \\ &= 44.415 - 30.963 \\ &= 13.452 \\ &= \underline{\underline{13.5^\circ}} \end{aligned}$$

$\triangle DCB$

$$\tan \theta = \frac{9}{15}$$

$$\theta = \tan^{-1}\left(\frac{9}{15}\right)$$

$$\theta = 30.963$$

$$13.5^\circ$$

(Total for Question 11 is 4 marks)

